

Tribhuvan University
Institute of Science and Technology

2077



BScCSIT Level/First Semester
Mathematics[MTH 112]
Calculus

Full Marks: 80
Pass Marks: 32
Time 3 Hrs.

Candidates are required to give their answers in their own words as far as practicable.

Group A ($10 \times 3 = 30$)

Attempt any **THREE** questions.

1. (a) If $f(x) = x^2$ then find $\frac{f(2+h)-f(2)}{h}$. ✓ [2]
- (b) (a) Dry air is moving upward. If the ground temperature is 20° and the temperature at a height of 1 km is 10°C , express the temperature T in $^\circ\text{C}$ as a function of the height h (in kilometers), assuming that a linear model is appropriate. (b) Draw the graph of the function in part (a). What does the slope represent? (c) What is the temperature at a height of 2 km ? [5]
- (c) Find the equation of the tangent to the parabola $y = x^2 + x + 1$ at $(0, 1)$. [3]
2. (a) A farmer has 2000 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area? [5]
- (b) Sketch the curve [3]

$$y = \frac{1}{x-3}$$

3. (a) Show that the $\int_1^\infty \frac{1}{x^2}$ converges and $\int_1^\infty \frac{1}{x}$ diverges. [2]

(b) If $f(x, y) = xy/(x^2 + y^2)$, does $f(x, y)$ exist, as $(x, y) \rightarrow (0, 0)$? [3]

(c) A particle moves in a straight line and has acceleration given by $a(t) = 6t^2 + 1$. Its initial velocity is $4m/sec$ and its initial displacement is $s(0) = 5cm$. Find its position function $s(t)$. [5]

4. (a) Evaluate [5]

$$\int_{-3}^2 \int_0^{\pi/2} (y + y^2 \cos x) dx dy$$

(b) Find the Maclaurin's series for $\cos x$ and prove that it represents $\cos x$ for all x . [5]

Group B ($10 \times 5 = 50$)

Attempt any TEN questions.

5. If $f(x) = x^2 - 1$, $g(x) = 2x + 1$, find $f \circ g$ and $g \circ f$ and domain of $f \circ g$.

6. Define continuity of a function at a point $x = a$. Show that the function $f(x) = \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

7. State Rolle's theorem and verify the Rolle's theorem for $f(x) = x^3 - x^2 - 6x + 2$ in $[0, 3]$.

8. Find the third approximation x_3 to the root of the equation $f(x) = x^3 - 2x - 7$, setting $x_1 = 2$.

9. Find the derivative of $r(t) = (1 + t^2)\mathbf{i} - te^{-t}\mathbf{j} + \sin 2t\mathbf{k}$ and find the unit tangent vector at $t = 0$.

10. Find the volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$.

11. Solve: $y'' + y' = 0$, $y(0) = 5$, $y(\pi/4) = 3$

12. Show that the series $\sum_{n=0}^{\infty} \frac{1}{1+n^2}$ converges.

13. Find a vector perpendicular to the plane that passes through the points: $P(1, 4, 6)$, $Q(-2, 5, -1)$ and $R(1, -1, 1)$.

14. Find the partial derivative of $f(x, y) = x^3 + 2x^2y^3 - 3y^2 + x + y$, at $(2, 1)$.

15. Find the local maximum and minimum values, saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.